XGBoost Analysis

David Alufi

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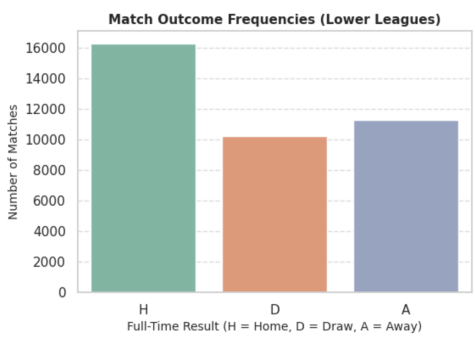
1 Introduction

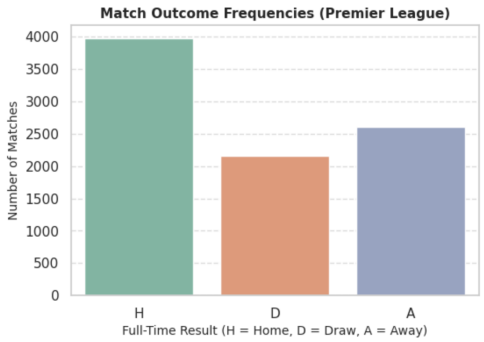
In this section, we are focusing on the impact of an XGBoost model. XGBoost is the baseline state-of-the-art framework that is built for accuracy and precision. I developed 2 XGBoost models to investigate our overall goal of fitting a classification model that tests the outcomes of Premier League matches using full time match statistics and opening odds from lower footballing leagues in England (Championship, League 1, and League 2). I felt it was interesting to develop 2 models simultaneously as I wanted to also focus on a predictive model using just opening betting odds and see where the differences lie. Our datasets measure all of these features from the 2003-04 season to the 2025-26 season. In total, we had 37,824 matches in our training datasets and 8,740 matches in our test dataset.

2 EDA

2.1 Match Frequencies

For the initial EDA, we will look at the match frequencies over the years in the lower leagues compared to the Premier League.





As we can see above, the frequencies are very similar with home wins having a 45.49% frequency in the Premier League and a 43.09% frequency in the lower leagues. We can use this to get our naive baseline: predicting Home Win for every match would give us a 45.49% frequency. Any model that

gave us a score lower than that would be a worse predictor than a trivial one. We also need a non-naive baseline, so we need to predict the outcome with the highest implied probability (this is the same as the lowest odds). If we derive this from the Bet365 opening odds (B365H, B365D, B365A), we get an accuracy of 46.44%. Yet again, any model that gave us a lower score wouldn't be useful.

2.2 Missingness

Another thing to consider was missingness, as we don't want any bias in the model. There are 16 columns without any missing values, one of those is FTR (standing for Full Time Result) and we want to remove that to prevent major target leakage as that is the thing we want to predict. That leaves 15 columns, split into 12 full time statistics and 3 odds markets. For our XGBoost models, we will be using these 15 columns for our classification.

3 Model Selection

3.1 Justification

XGBoost was chosen for a number of reasons: by sequentially training decision trees to learn from the errors of previous trees, gradient boosting can outperform individual decision trees or basic linear models for tabulated data. In addition, the non-linear relationship between match statistics and market odds favour tree-based ensemble methods as they do not need any feature scaling.

3.2 Training Framework

The model is exclusively trained on the season-by-season history of the combination of match statistics and betting odds from the lower leagues. This gives the algorithm a large volume of data to learn patterns. The test dataset will remain unseen until the end in order to prevent data leakage.

3.3 Model Choices

Since there were 15 complete columns, it made sense to remove the others so that we could have the model with the least imputation errors. We can guarantee a consistent and reliable feature space, whilst also trying to prevent leakage by removing the full time outcomes. However, the 15-feature model still incorporates full time statistics as inputs. When trying to predict an unseen Premier League match, these full time statistics introduce temporal leakage since the model accidentally accesses information from the future. While using these statistics remains valid for a classification model aimed at understanding how performance metrics relate to match outcomes, it isn't suitable for pre-match forecasting. To accurately attempt to predict future games before they begin, we can also look at pre-match inputs in solitary. This motivates my second choice - an odds only model. Since betting odds are fully finalised before a match begins, this setup provides a temporally clean baseline that reflects true pre-match predictive conditions.

3.4 Hyperparameter Configuration

To maintain a standard baseline, the XGBoost classifier is initialised using Python's standard default hyperparameters. No optimsation was done.

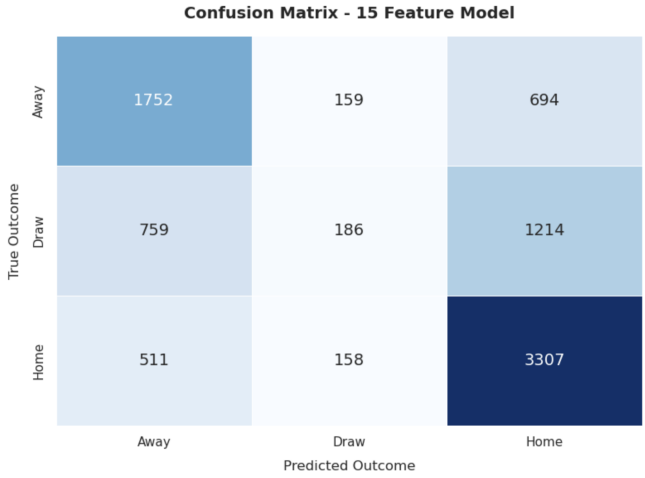
3.5 Evaluation Metrics

The metric chosen is multi-class log-loss. Log-loss measures the penalty of a model's predicted probabilities compared to the actual outcomes, heavily punishing models that are confidently wrong. It evaluates how well calibrated your probability distributions are across multiple classes (Home, Draw, Away), providing a much more rigorous comparison between the 15-feature and odds only models than just accuracy score. The log-loss is not just limited to this, we can also use it to compare ensemble methods, e.g. XGBoost vs Random Forest. The closer the log-loss score is to 0, the closer the model is to assigning a 100% probability to every correct match outcome.

4 Evaluating XGBoost

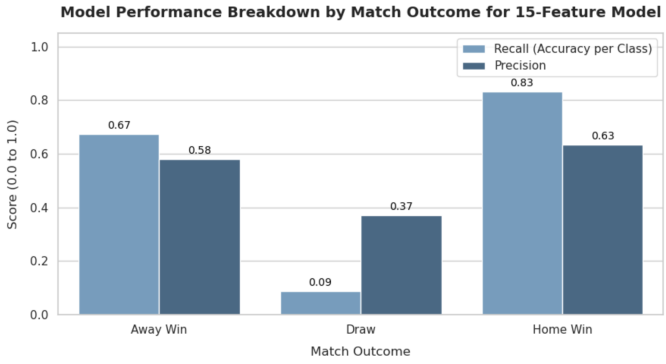
4.1 Model Performance

Firstly, the 15-feature model achieved a classification accuracy of 60.01% on the unseen Premier League dataset, with a multi-class log-loss score of 0.8646. I also wanted to evaluate the performance across individual outcomes (Home, Draw, Away) and computed a confusion matrix.

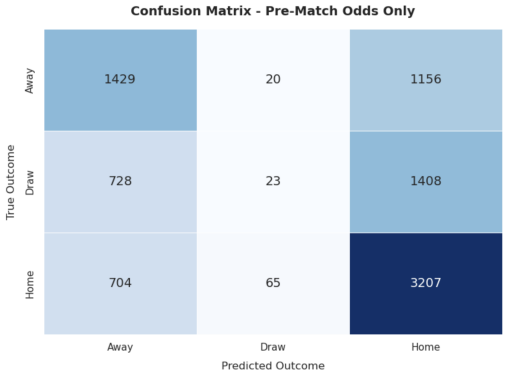


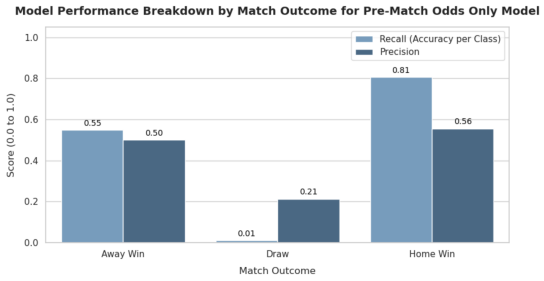
As seen, the model has a clear disparity in accuracy depending on the outcome. It is really good at predicting Home Wins, and has decent predictions on Away Wins. It struggles with Draws a lot, frequently misclassifying Draws as either Home or Away Wins, due to the high variance and statistical noise typical of Draws. In addition, precision and recall metrics were evaluated for all three outcomes. Precision measures the reliability of the model's positive predictions, e.g., out of all matches the model predicted as a given outcome, what proportion were actually correct. Recall measures sensitivity, representing the proportion of actual outcomes in the test dataset that the

model successfully identified. The model accurately predicted 83% and 67% of all actual Home and Away Wins respectively, whilst having a sharp decline in the accuracy in Draws.



The pre-match odds only model did not perform as well, having a classification accuracy of 53.31% with a multi-class log-loss score of 0.9852. This model has a similar disparity to the last. It is also very strong at predicting Home Wins, but differs in its ability to predict Away Wins. Since pre-match odds can favour the home side more often than not, this model heavily favours Home results over Draws or Away results. It very rarely predicts Draws correctly, which is another flaw to the odds only model. To no surprise, this model has a strong recall score towards Home Win, correctly predicting 81% of the total. By comparing the two models, the precision and recall scores aren't too far apart, however we would always choose the 15-feature model as it has a lower log-loss score.

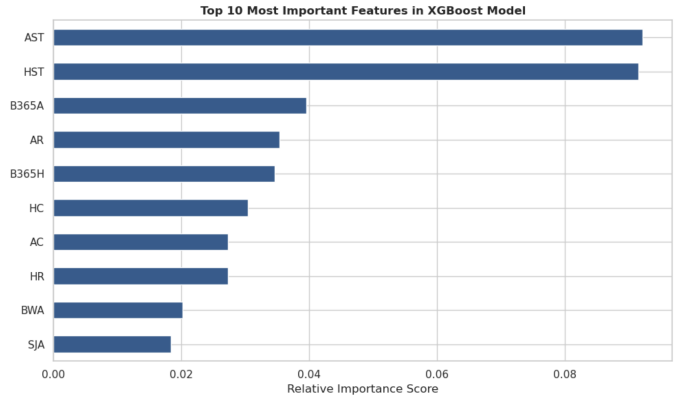




4.2 Feature Importance

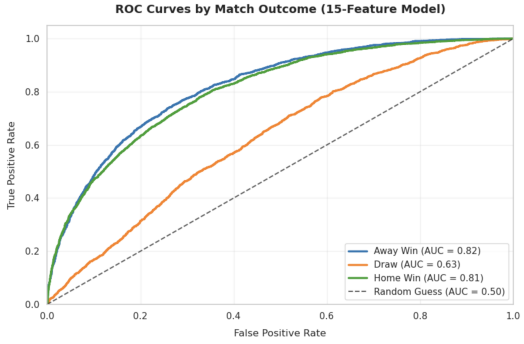
Here, we will go over the features that have the highest importance towards the XGBoost Model. It uses a Relative Importance Score, calculated using Gain, which measures the actual importance each feature brings to the model. Unsurprisingly, away and home shots on target (AST & HST) are the two most important. Shots on target represent opportunities to score goals, which make them

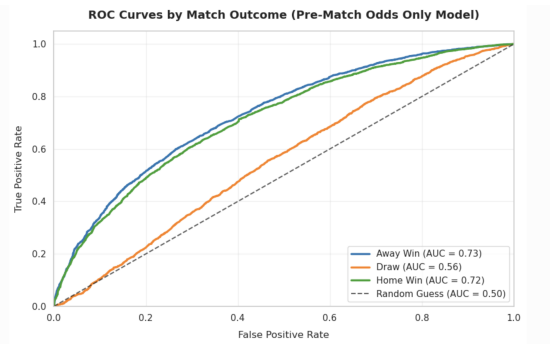
the most informative signal for separating between outcomes. AR and HR represent red cards for both teams; receiving a red card gives the other team a huge physical advantage which has huge implications on final match outcomes. For this reason, I don't find it surprising that both of these features rank inside the top 10. Most of the other features are bookmaker odds. An interesting conclusion to draw from this is most of the bookmaker odds in the top 10 are away odds. From our EDA, we see that there tends to be a home advantage; so to counteract this baseline home bias, extra emphasis must be placed on some bookmaker away odds. To complete the top 10, HC and AC represent corners for both teams; whilst not as important as shots, corners represent sustained pressure and territorial dominance. Overall, we can see there are a bunch of different features that the XGBoost model finds important. It is good to note this is a general case, our 15-feature model would not have used the bookmaker odds other than Bet365.



4.3 ROC Curves

An ROC curve evaluates how well our model distinguishes between different outcomes across different probability thresholds. We look at the area under the curve (AUC) to represent the probability that your model ranks a randomly chosen correct outcome higher than a randomly chosen incorrect outcome. We want the area to be as close to 1 as that gives a 100% probability.





We can see that the broad shape of the curves are similar, but the 15-feature model does outperform the pre-match odds only in all three outcomes. Both are barely above 0.5 for Draws, reinforcing the difficulty in ranking a randomly chosen correct outcome higher than an incorrect one. In both models, the Home Win and Away Win curves remain very close to one another, with Away Win achieving a minor 0.01 higher AUC score. This small difference is statistically negligible, indicating that both models discriminate Home and Away wins with virtually equal effectiveness across all probability thresholds.

5 Conclusion

Scoring a 60.01% classification accuracy with a 0.8646 multi-class log-loss score for the 15-feature model ends up being better than the baseline established in the EDA as well as the odds only model. The model favoured features such as shots on target and red cards which would help suggest why the pre-match did not perform as well. Both struggled with predicting Draws and may not improve unless further match statistics become popularised like other sports such as the NBA.